Multi-Model Classification for Customer Churn Prediction

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Set working Directory

```
# setwd("C:/Users/user/Desktop/Professional DS Mule/R")
```

**Task 1 : Data Import and Exploration

Load required libraries

```
library(tidyverse)      # Data Manipulation
```

```
## ── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
## ✔ dplyr     1.2.0     ✔ readr     2.2.0
## ✔ forcats   1.0.1     ✔ stringr   1.6.0
## ✔ ggplot2   4.0.3     ✔ tibble    3.3.1
## ✔ lubridate 1.9.5     ✔ tidyr     1.3.2
## ✔ purrr     1.2.1
## ── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()    masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(caret)          # Unified ML Interface
```

```
## Loading required package: lattice
##
## Attaching package: 'caret'
##
## The following object is masked from 'package:purrr':
##
##     lift
```

```
library(pROC)           # ROC Analysis
```

```
## Type 'citation("pROC")' for a citation.
##
## Attaching package: 'pROC'
##
## The following objects are masked from 'package:stats':
##
##     cov, smooth, var
```

```
library(e1071)          # SVM
```

```
##
## Attaching package: 'e1071'
##
## The following object is masked from 'package:ggplot2':
##
##     element
```

```
library(class)          # KNN
library(randomForest)    # Random Forest
```

```
## randomForest 4.7-1.2
## Type rfNews() to see new features/changes/bug fixes.
##
## Attaching package: 'randomForest'
##
## The following object is masked from 'package:dplyr':
##
##     combine
##
## The following object is masked from 'package:ggplot2':
##
##     margin
```

```
library(psych)
```

```
##
## Attaching package: 'psych'
##
## The following object is masked from 'package:randomForest':
##
##     outlier
##
## The following objects are masked from 'package:ggplot2':
##
##     %+%, alpha
```

```
 load("C:/Users/user/Desktop/DATA SCIENTIST MULE/R/ASSIGNMENTS/business_churn_data.RData")
```

```
#Load the data
data <-business_data
head(data)
```

```
##   CustomerID Age Gender TenureMonths MonthlyCharges TotalCharges   ContractType
## 1      10001  44   Male            4          65.88       267.08 Month-to-month
## 2      10002  25   Male            6          61.23       378.56       One-year
## 3      10003  40   Male            1         121.37       118.39 Month-to-month
## 4      10004  37   Male            2          57.57       118.92       Two-year
## 5      10005  30   Male           36          77.26      2760.74 Month-to-month
## 6      10006  18   Male           47          40.61      1971.22       One-year
##      PaymentMethod InternetService OnlineSecurity TechSupport StreamingTV
## 1     Mailed check     Fiber optic             No          No          No
## 2    Bank transfer             DSL            Yes          No         Yes
## 3    Bank transfer     Fiber optic            Yes          No         Yes
## 4 Electronic check     Fiber optic             No          No         Yes
## 5 Electronic check             DSL             No          No         Yes
## 6    Bank transfer     Fiber optic             No          No          No
##   StreamingMovies NumSupportTickets SatisfactionScore LoyaltyProgram Churn
## 1              No                 4                 6            Yes     1
## 2              No                 3                10             No     0
## 3             Yes                 8                 5             No     1
## 4             Yes                 4                 5             No     0
## 5              No                 4                 8             No     0
## 6              No                 1                 7             No     0
```

Explore the dataset

```
str(data)
```

```
## 'data.frame':    2560 obs. of  17 variables:
##  $ CustomerID       : int  10001 10002 10003 10004 10005 10006 10007 10008 10009 10010 ...
##  $ Age              : num  44 25 40 37 30 18 29 26 39 32 ...
##  $ Gender           : Factor w/ 2 levels "Female","Male": 2 2 2 2 2 2 2 1 2 1 ...
##  $ TenureMonths     : num  4 6 1 2 36 47 51 37 25 20 ...
##  $ MonthlyCharges   : num  65.9 61.2 121.4 57.6 77.3 ...
##  $ TotalCharges     : num  267 379 118 119 2761 ...
##  $ ContractType     : Factor w/ 3 levels "Month-to-month",..: 1 2 1 3 1 2 1 2 1 1 ...
##  $ PaymentMethod    : Factor w/ 4 levels "Bank transfer",..: 4 1 1 3 3 1 4 3 4 4 ...
##  $ InternetService  : Factor w/ 3 levels "DSL","Fiber optic",..: 2 1 2 2 1 2 2 1 1 2 ...
##  $ OnlineSecurity   : Factor w/ 3 levels "No","No internet",..: 1 3 3 1 1 1 3 1 3 1 ...
##  $ TechSupport      : Factor w/ 3 levels "No","No internet",..: 1 1 1 1 1 1 3 1 1 3 ...
##  $ StreamingTV      : Factor w/ 3 levels "No","No internet",..: 1 3 3 3 3 1 3 1 1 1 ...
##  $ StreamingMovies  : Factor w/ 3 levels "No","No internet",..: 1 1 3 3 1 1 1 1 1 3 ...
##  $ NumSupportTickets: num  4 3 8 4 4 1 5 5 2 10 ...
##  $ SatisfactionScore: num  6 10 5 5 8 7 5 10 8 5 ...
##  $ LoyaltyProgram   : Factor w/ 2 levels "No","Yes": 2 1 1 1 1 1 1 2 2 2 ...
##  $ Churn            : int  1 0 1 0 0 0 1 0 0 1 ...
```

```
describe(data)
```

```
##                   vars    n     mean      sd   median  trimmed     mad      min
## CustomerID           1 2560 11280.50  739.15 11280.50 11280.50  948.86 10001.00
## Age                  2 2560    38.26   11.31    38.00    38.04   11.86    18.00
## Gender*              3 2560     1.53    0.50     2.00     1.53    0.00     1.00
## TenureMonths         4 2560    25.92   20.73    20.00    23.90   20.76     1.00
## MonthlyCharges       5 2560    83.21   34.93    81.98    82.38   35.39    20.00
## TotalCharges         6 2560  2163.35 2084.57  1386.59  1829.37 1520.42    19.29
## ContractType*        7 2560     1.66    0.80     1.00     1.57    0.00     1.00
## PaymentMethod*       8 2560     2.49    1.06     3.00     2.48    1.48     1.00
## InternetService*     9 2560     1.85    0.73     2.00     1.81    1.48     1.00
## OnlineSecurity*     10 2560     1.90    0.89     2.00     1.88    1.48     1.00
## TechSupport*        11 2560     1.85    0.88     2.00     1.82    1.48     1.00
## StreamingTV*        12 2560     2.06    0.89     2.00     2.08    1.48     1.00
## StreamingMovies*    13 2560     2.00    0.89     2.00     2.00    1.48     1.00
## NumSupportTickets   14 2560     4.10    2.02     4.00     4.02    1.48     0.00
## SatisfactionScore   15 2560     6.81    2.11     7.00     6.89    2.97     1.00
## LoyaltyProgram*     16 2560     1.36    0.48     1.00     1.33    0.00     1.00
## Churn               17 2560     0.32    0.47     0.00     0.27    0.00     0.00
##                        max    range  skew kurtosis    se
## CustomerID        12560.00  2559.00  0.00    -1.20 14.61
## Age                  75.00    57.00  0.20    -0.41  0.22
## Gender*               2.00     1.00 -0.11    -1.99  0.01
## TenureMonths         72.00    71.00  0.65    -0.86  0.41
## MonthlyCharges      200.00   180.00  0.26    -0.22  0.69
## TotalCharges      12605.68 12586.39  1.39     1.77 41.20
## ContractType*         3.00     2.00  0.69    -1.09  0.02
## PaymentMethod*        4.00     3.00 -0.11    -1.23  0.02
## InternetService*      3.00     2.00  0.24    -1.09  0.01
## OnlineSecurity*       3.00     2.00  0.19    -1.71  0.02
## TechSupport*          3.00     2.00  0.29    -1.65  0.02
## StreamingTV*          3.00     2.00 -0.12    -1.73  0.02
## StreamingMovies*      3.00     2.00  0.00    -1.75  0.02
## NumSupportTickets    12.00    12.00  0.45     0.12  0.04
## SatisfactionScore    10.00     9.00 -0.34    -0.46  0.04
## LoyaltyProgram*       2.00     1.00  0.58    -1.67  0.01
## Churn                 1.00     1.00  0.79    -1.38  0.01
```

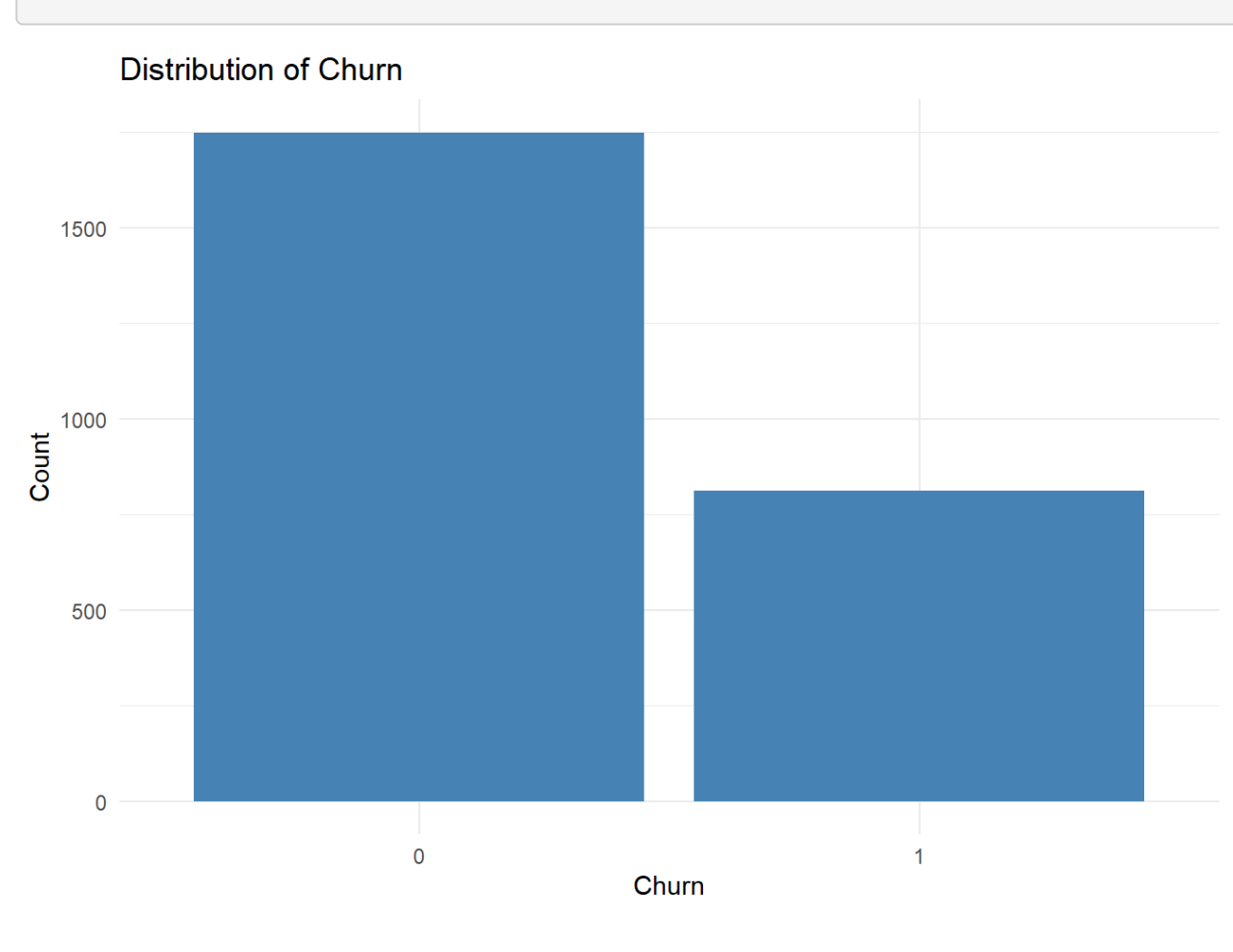
Check for missing values

```
# Count missing values per column
missing_count <- colSums(is.na(data))
missing_count
```

```
##        CustomerID               Age            Gender      TenureMonths
##                 0                 0                 0                 0
##    MonthlyCharges      TotalCharges      ContractType     PaymentMethod
##                 0                 0                 0                 0
##   InternetService    OnlineSecurity       TechSupport       StreamingTV
##                 0                 0                 0                 0
##   StreamingMovies NumSupportTickets SatisfactionScore    LoyaltyProgram
##                 0                 0                 0                 0
##             Churn
##                 0
```

Visualize the distribution of the target variable Churn

```
ggplot(data, aes(x = factor(Churn))) +
  geom_bar(fill = "steelblue") +
  labs(title = "Distribution of Churn", x = "Churn", y = "Count") +
  theme_minimal()
```



Class Imbalance Ratio

```
# Frequency table for Churn variable
churn_table <- table(data$Churn)
# Display the frequency table
churn_table
```

```
##
##    0    1
## 1749  811
```

```
# Calculate the imbalance ratio
imbalance_ratio <- churn_table[1] / churn_table[2]
imbalance_ratio
```

```
##        0
## 2.156597
```

Task 2 : Data Preprocessing

Convert target variable to factor

```
data$Churn <- factor(data$Churn, levels = c(0, 1), labels = c("No", "Yes"))
```

Convert Categorical variables to factors

```
categorical_vars <- c("Gender","ContractType","PaymentMethod","InternetService","TechSupport","OnlineSecurity","L
oyaltyProgram","StreamingTV","StreamingMovies")
data[categorical_vars] <- lapply(data[categorical_vars], as.factor)
```

Task 3 : Data Partitioning

Split the dataset into training and testing sets

```
set.seed(2026)  # For reproducibility
#Stratified sampling to maintain class distribution
train_index <- createDataPartition(data$Churn, p = 0.7, list = FALSE)
train_data <- data[train_index, ]
test_data <- data[-train_index, ]
```

Check class distribution in training and testing sets

```
table(train_data$Churn)
```

```
##
##   No  Yes
## 1225  568
```

```
table(test_data$Churn)
```

```
##
##  No Yes
## 524 243
```

Apply scaling to numeric features

```
numeric_vars <- sapply(train_data, is.numeric)
prepocessing_model <- preProcess(train_data[, numeric_vars], method = c("center", "scale"))
train_data_scaled <- predict(prepocessing_model, train_data)
test_data_scaled <- predict(prepocessing_model, test_data)
```

Task 4 : Train Control Scheme

Define train control for cross-validation

```
ctrl <- trainControl(
  method = "cv",           # Cross-validation
  number = 10,            # Number of folds
  classProbs = TRUE,      # Needed for AUC calculation
  summaryFunction = twoClassSummary,  #ROC,Sensitivity,Specificity
  savePredictions = "final" # Save predictions for the best model
)
```

Task 5 :Model Training - Logistic Regression

```
set.seed(2026)
logistic_model <- train(Churn ~ ., data = train_data, method = "glm", family = "binomial", trControl = ctrl, metr
ic = "ROC")
```

```
# view coeeficients of the logistic regression model
summary(logistic_model$finalModel)$coefficients
```

```
##                                          Estimate   Std. Error    z value
## (Intercept)                          0.9610780269 1.124920e+00  0.8543521
## CustomerID                          -0.0000500942 9.054408e-05 -0.5532576
## Age                                 -0.0025902126 6.006181e-03 -0.4312579
## GenderMale                          -0.1429482323 1.330452e-01 -1.0744340
## TenureMonths                        -0.0594288896 1.254112e-02 -4.7387236
## MonthlyCharges                       0.0254556597 3.464408e-03  7.3477664
## TotalCharges                         0.0002084686 1.199272e-04  1.7382932
## `ContractTypeOne-year`              -0.6827055856 1.579132e-01 -4.3232952
## `ContractTypeTwo-year`              -1.9134575681 2.299541e-01 -8.3210426
## `PaymentMethodCredit card`          -0.0236244920 1.995560e-01 -0.1183853
## `PaymentMethodElectronic check`  0.2848962449 1.763715e-01  1.6153191
## `PaymentMethodMailed check`     -0.2087291651 2.029701e-01 -1.0283740
## `InternetServiceFiber optic`     0.2689711017 1.531457e-01  1.7563089
## InternetServiceNo                   -1.0095558903 2.411435e-01 -4.1865358
## OnlineSecurityYes                   -0.2538240427 1.473571e-01 -1.7225100
## TechSupportYes                       0.1295329779 1.484078e-01  0.8728178
## StreamingTVYes                       0.0331763842 1.460285e-01  0.2271911
## StreamingMoviesYes                   0.0595729367 1.456031e-01  0.4091462
## NumSupportTickets                    0.1125345711 3.732160e-02  3.0152662
## SatisfactionScore                   -0.3440059963 3.442267e-02 -9.9935886
## LoyaltyProgramYes                   -0.2637708914 1.387228e-01 -1.9014246
##                                     Pr(>|z|)
## (Intercept)                         3.929099e-01
## CustomerID                          5.800870e-01
## Age                                 6.662809e-01
## GenderMale                          2.826282e-01
## TenureMonths                        2.150687e-06
## MonthlyCharges                      2.015463e-13
## TotalCharges                        8.215916e-02
## `ContractTypeOne-year`              1.537158e-05
## `ContractTypeTwo-year`              8.719290e-17
## `PaymentMethodCredit card`          9.057624e-01
## `PaymentMethodElectronic check` 1.062416e-01
## `PaymentMethodMailed check`     3.037739e-01
## `InternetServiceFiber optic`    7.903567e-02
## InternetServiceNo                   2.832443e-05
## OnlineSecurityYes                   8.497717e-02
## TechSupportYes                      3.827624e-01
## StreamingTVYes                      8.202751e-01
## StreamingMoviesYes                  6.824324e-01
## NumSupportTickets                   2.567538e-03
## SatisfactionScore                   1.625868e-23
## LoyaltyProgramYes                   5.724643e-02
```

```
# Calculate odds ratios using tab_model function from sjPlot package
library(sjPlot)
```

```
##
## Attaching package: 'sjPlot'
```

```
## The following object is masked from 'package:ggplot2':
##
##     set_theme
```

```
tab_model(logistic_model$finalModel, show.ci = FALSE, show.se = FALSE, show.stat = FALSE, show.p = FALSE, show.ai
c = FALSE)
```

```
## Model matrix is rank deficient. Parameters `OnlineSecurityNo internet,
##   TechSupportNo internet, StreamingTVNo internet, StreamingMoviesNo
##   internet` were not estimable.
```

	Dependent variable
Predictors	*Odds Ratios*
(Intercept)	2.61
CustomerID	1.00
Age	1.00
GenderMale	0.87
TenureMonths	0.94
MonthlyCharges	1.03
TotalCharges	1.00
ContractTypeOne-year	0.51
ContractTypeTwo-year	0.15
PaymentMethodCredit card	0.98
PaymentMethodElectronic check	1.33
PaymentMethodMailed check	0.81
InternetServiceFiber optic	1.31
InternetServiceNo	0.36
OnlineSecurityYes	0.78
TechSupportYes	1.14
StreamingTVYes	1.03
StreamingMoviesYes	1.06
NumSupportTickets	1.12
SatisfactionScore	0.71
LoyaltyProgramYes	0.77
Observations	1793
R^2 Tjur	0.408

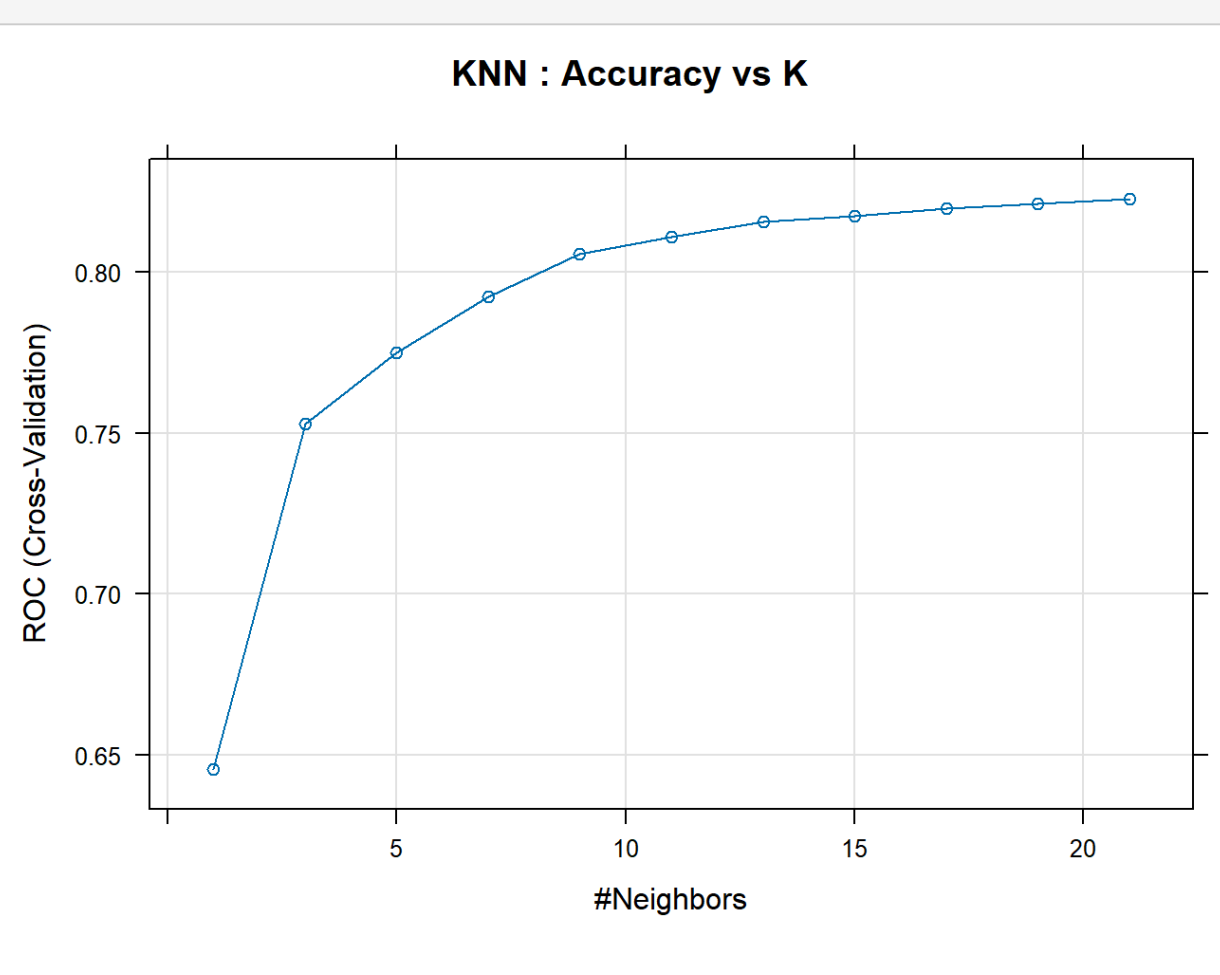
Task 6 : Model Training -KNN

```
set.seed(2026)
knn_model <- train(Churn ~ ., data = train_data_scaled, method = "knn", trControl = ctrl, metric = "ROC",
                   tuneGrid = expand.grid(k = seq(1, 21, by = 2)))
```

```
# View the optimal k value for KNN
knn_model$bestTune
```

```
##     k
## 11 21
```

```
# Plot KNN performance across different k values
plot(knn_model,main = "KNN : Accuracy vs K")
```



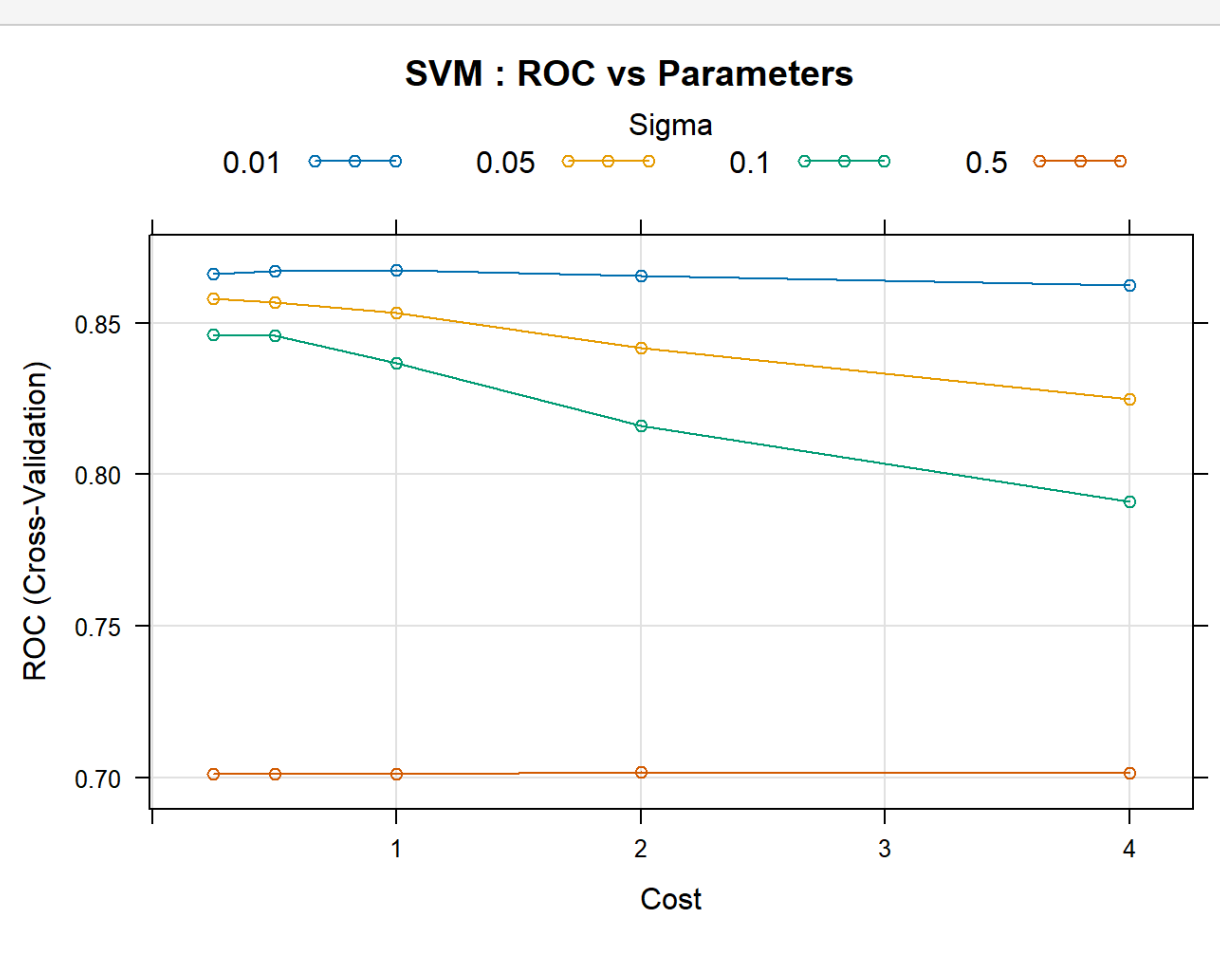
.. ## **Task 7 : Model Training - SVM**

```
set.seed(2026)
svm_model <- train(Churn ~ ., data = train_data_scaled, method = "svmRadial", trControl = ctrl, metric = "ROC",
                   tuneGrid = expand.grid(C = c(0.25, 0.5, 1,2,4), sigma = c(0.01,0.05, 0.1, 0.5)))
```

```
# View the best parameters for SVM
svm_model$bestTune
```

```
##   sigma C
## 9  0.01 1
```

```
#Plot Tuning results for SVM
plot(svm_model, main = "SVM : ROC vs Parameters")
```



Task 8 : Model Training - Random Forest

```
rf_model <- train(Churn ~ ., data = train_data, method = "rf",  trControl = ctrl, metric = "ROC",
                  tuneGrid = expand.grid(mtry = c(2, 4, 6, 8, 10)))
```

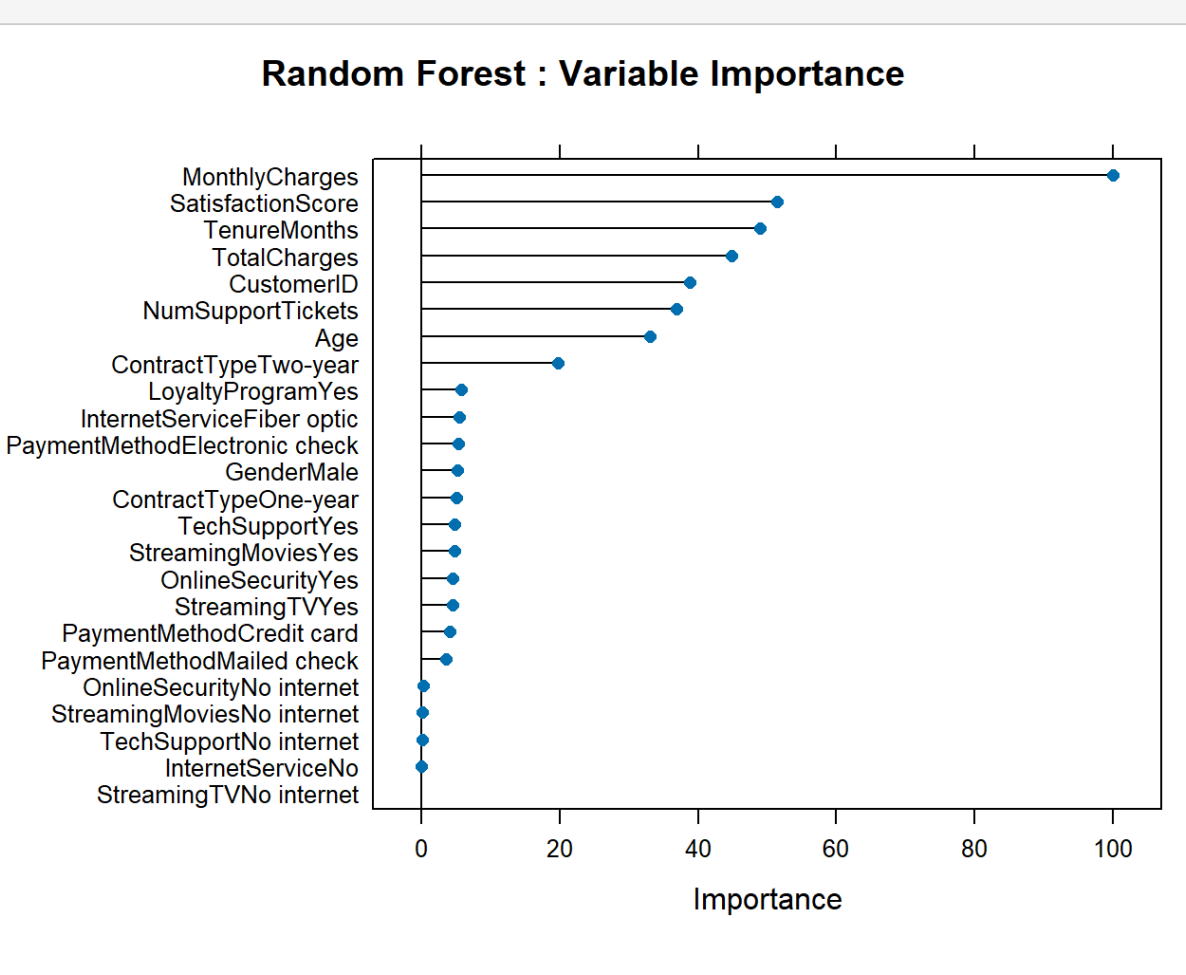
```
# View the best mtry value for Random Forest
rf_model$bestTune
```

```
##   mtry
## 2    4
```

```
#Plot variable importance for Random Forest
varImp(rf_model)
```

```
## rf variable importance
##
##   only 20 most important variables shown (out of 24)
##
##                                Overall
## MonthlyCharges                100.0000
## SatisfactionScore              51.4456
## TenureMonths                   49.0251
## TotalCharges                   44.7907
## CustomerID                     38.8799
## NumSupportTickets              36.9729
## Age                            33.1172
## ContractTypeTwo-year           19.8341
## LoyaltyProgramYes               5.7266
## InternetServiceFiber optic      5.5051
## PaymentMethodElectronic check   5.3529
## GenderMale                      5.2462
## ContractTypeOne-year            5.1029
## TechSupportYes                  4.8855
## StreamingMoviesYes              4.8680
## OnlineSecurityYes               4.6196
## StreamingTVYes                  4.5691
## PaymentMethodCredit card        4.0954
## PaymentMethodMailed check       3.6075
## OnlineSecurityNo internet       0.3246
```

```
plot(varImp(rf_model), main = "Random Forest : Variable Importance")
```



Task 9 : Model Evaluation and Comparison

Predictions for all models

```
logistic_preds <- predict(logistic_model, test_data, type = "prob")[, "Yes"]
knn_preds <- predict(knn_model, test_data_scaled, type = "prob")[, "Yes"]
svm_preds <- predict(svm_model, test_data_scaled, type = "prob")[, "Yes"]
rf_preds <- predict(rf_model, test_data, type = "prob")[, "Yes"]
```

Confusion Matrices for each Model

```
# Confusion Matrix for Logistic Regression
logistic_cm <- confusionMatrix(predict(logistic_model, test_data), test_data$Churn,positive = "Yes")
logistic_cm
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction  No Yes
##        No  457  87
##        Yes  67 156
##
##                Accuracy : 0.7992
##                  95% CI : (0.7691, 0.827)
##     No Information Rate : 0.6832
##     P-Value [Acc > NIR] : 4.275e-13
##
##                   Kappa : 0.5257
##
##  Mcnemar's Test P-Value : 0.1258
##
##             Sensitivity : 0.6420
##             Specificity : 0.8721
##          Pos Pred Value : 0.6996
##          Neg Pred Value : 0.8401
##              Prevalence : 0.3168
##          Detection Rate : 0.2034
##    Detection Prevalence : 0.2907
##       Balanced Accuracy : 0.7571
##
##        'Positive' Class : Yes
##
```

```
# Confusion Matrix for KNN
knn_cm <- confusionMatrix(predict(knn_model, test_data_scaled), test_data$Churn,positive = "Yes")
knn_cm
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction  No Yes
##        No  468 112
##        Yes  56 131
##
##                Accuracy : 0.781
##                  95% CI : (0.75, 0.8098)
##     No Information Rate : 0.6832
##     P-Value [Acc > NIR] : 1.168e-09
##
##                   Kappa : 0.4607
##
##  Mcnemar's Test P-Value : 2.202e-05
##
##             Sensitivity : 0.5391
##             Specificity : 0.8931
##          Pos Pred Value : 0.7005
##          Neg Pred Value : 0.8069
##              Prevalence : 0.3168
##          Detection Rate : 0.1708
##    Detection Prevalence : 0.2438
##       Balanced Accuracy : 0.7161
##
##        'Positive' Class : Yes
##
```

```
# Confusion Matrix for SVM
svm_cm <- confusionMatrix(predict(svm_model, test_data_scaled), test_data$Churn,positive = "Yes")
svm_cm
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction  No Yes
##        No  464  85
##        Yes  60 158
##
##                Accuracy : 0.811
##                  95% CI : (0.7814, 0.8381)
##     No Information Rate : 0.6832
##     P-Value [Acc > NIR] : 1.163e-15
##
##                   Kappa : 0.5509
##
##  Mcnemar's Test P-Value : 0.04625
##
##             Sensitivity : 0.6502
##             Specificity : 0.8855
##          Pos Pred Value : 0.7248
##          Neg Pred Value : 0.8452
##              Prevalence : 0.3168
##          Detection Rate : 0.2060
##    Detection Prevalence : 0.2842
##       Balanced Accuracy : 0.7679
##
##        'Positive' Class : Yes
##
```

```
# Confusion Matrix for Random Forest
rf_cm <- confusionMatrix(predict(rf_model, test_data), test_data$Churn,positive = "Yes")
rf_cm
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction  No Yes
##        No  469 107
##        Yes  55 136
##
##                Accuracy : 0.7888
##                  95% CI : (0.7582, 0.8172)
##     No Information Rate : 0.6832
##     P-Value [Acc > NIR] : 4.730e-11
##
##                   Kappa : 0.4824
##
##  Mcnemar's Test P-Value : 6.151e-05
##
##             Sensitivity : 0.5597
##             Specificity : 0.8950
##          Pos Pred Value : 0.7120
##          Neg Pred Value : 0.8142
##              Prevalence : 0.3168
##          Detection Rate : 0.1773
##    Detection Prevalence : 0.2490
##       Balanced Accuracy : 0.7274
##
##        'Positive' Class : Yes
##
```

ROC Curves and AUC for each Model

```
# All ROC curves in one plot
roc_logistic <- roc(test_data$Churn, logistic_preds)
```

```
## Setting levels: control = No, case = Yes
```

```
## Setting direction: controls < cases
```

```
roc_knn <- roc(test_data$Churn, knn_preds)
```

```
## Setting levels: control = No, case = Yes
## Setting direction: controls < cases
```

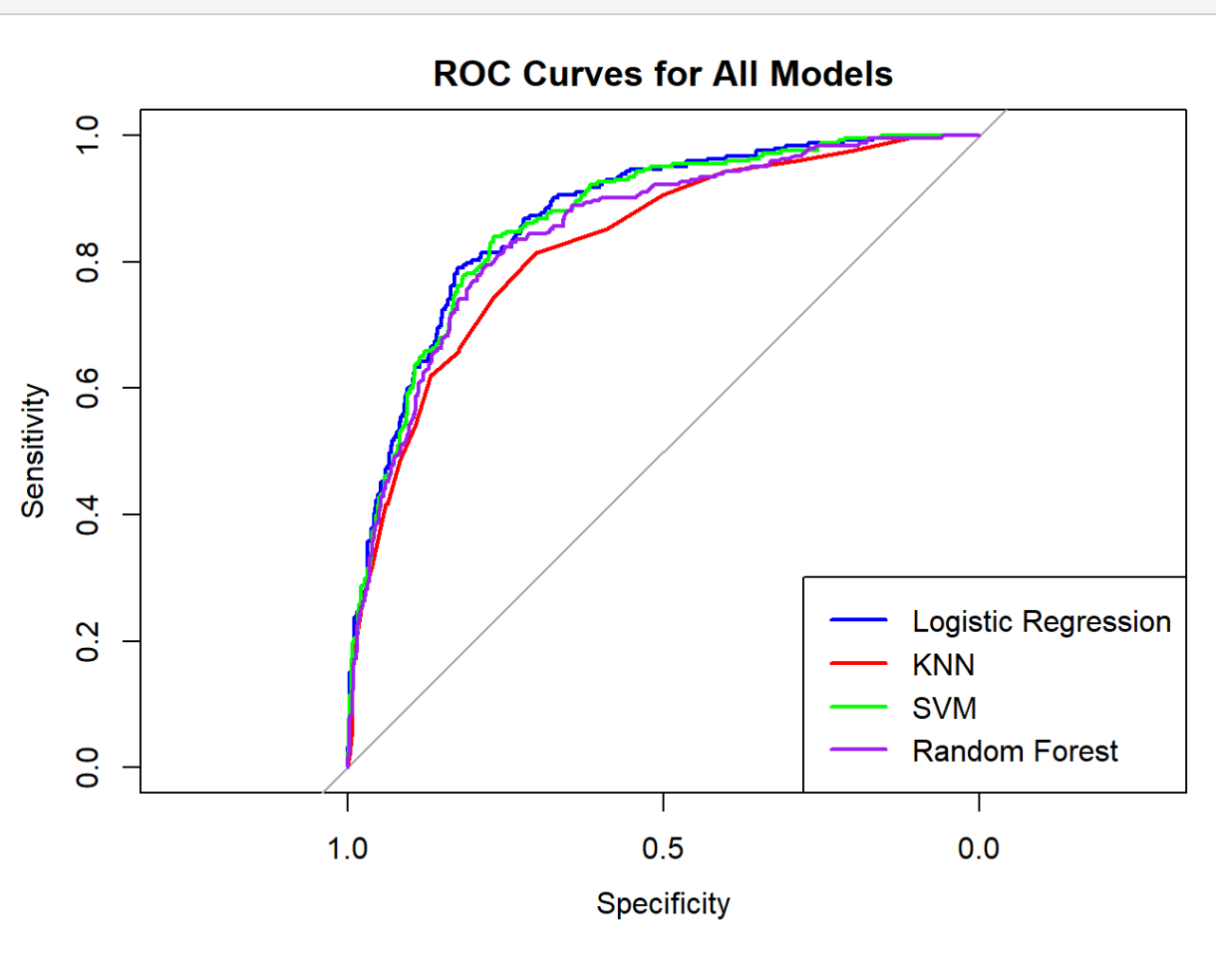
```
roc_svm <- roc(test_data$Churn, svm_preds)
```

```
## Setting levels: control = No, case = Yes
## Setting direction: controls < cases
```

```
roc_rf <- roc(test_data$Churn, rf_preds)
```

```
## Setting levels: control = No, case = Yes
## Setting direction: controls < cases
```

```
plot(roc_logistic, col = "blue", main = "ROC Curves for All Models")
plot(roc_knn, col = "red", add = TRUE)
plot(roc_svm, col = "green", add = TRUE)
plot(roc_rf, col = "purple", add = TRUE)
legend("bottomright", legend = c("Logistic Regression", "KNN", "SVM", "Random Forest"), col = c("blue", "red", "g
reen", "purple"), lwd = 2)
```



AUC Comparison

```
auc_values <- data.frame(
  Model = c("Logistic Regression", "KNN", "SVM", "Random Forest"),
  AUC = c(auc(roc_logistic), auc(roc_knn), auc(roc_svm), auc(roc_rf))
)
print(auc_values)
```

```
##                 Model       AUC
## 1 Logistic Regression 0.8698599
## 2                 KNN 0.8268856
## 3                 SVM 0.8646844
## 4       Random Forest 0.8487262
```